t2-w09 — condition C1

Simplification of Algebraic Terms — Simplification of algebraic expressions: order of operations, brackets and word problems

> The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	**present**
Assessment	the questions the lesson was checked with	prompt	**present**
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Simplification of Algebraic Terms

`subtopic`

Simplification of algebraic expressions: order of operations, brackets and word problems

`term`

Second Term

`week`

9

`lesson_objectives`

• carry out order of operation in algebra
• insert/remove bracket and simplify expression
• solve word problems involving algebraic expression

`entry_behaviour`

• Addition and subtraction of algebraic terms
• Multiplication and division of algebraic terms

`instructional_materials`

• Chart showing algebraic expression.

Objective 1 — carry out order of operation in algebra

`objective_text`

carry out order of operation in algebra

`current_step`

1

`instructional_events`

• provide_learning_guidance
• elicit_performance
• provide_feedback

`student_tasks`

• 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$

`assessment_points`

• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$
• Simplify $7 \times 3a - (3a + 5a) \times 2$

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — insert/remove bracket and simplify expression

objective_text

insert/remove bracket and simplify expression

current_step

2

instructional_events

• provide_learning_guidance
• elicit_performance
• provide_feedback

student_tasks

• (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$

assessment_points

• Simplify $12 - (7 - 2f) - (3 - f)$
• Simplify $(8a - 5) + (14 - 3a)$

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, activities, source_evidence, source_figures, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 3 — solve word problems involving algebraic expression

objective_text

solve word problems involving algebraic expression

current_step

3

instructional_events

• provide_learning_guidance
• elicit_performance
• provide_feedback

student_tasks

• 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.

assessment_points

• A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, activities, source_evidence, source_figures, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Using the Order of Operations to Simplify Algebraic Expressions

> EXPLANATION
> When you simplify an algebraic expression you must follow the same order of operations you use with numbers: Parentheses (brackets) – work inside them first, Exponents, Multiplication and Division (from left to right), Addition and Subtraction (from left to right).
>
> In algebra the symbols represent numbers, so the rules work exactly the same. Remember to treat each term as a whole when you multiply or divide – for example, $3x$ means $3 \times x$.

> WORKED_EXAMPLE
> **Worked example**
>
> Simplify $4n + 3n \times 10$.
>
> 1. Look for any brackets – there are none.
> 2. Do any multiplication or division from left to right.
> $3n \times 10 = 30n$.
> 3. Now add the remaining terms: $4n + 30n = 34n$.
>
> **Result:** $34n$.
>
> ---
>
> **Another example**
>
> Simplify $32 + 8x \div 2$.
>
> 1. No brackets.
> 2. Perform the division first: $8x \div 2 = 4x$.
> 3. Add the remaining terms: $32 + 4x$ (cannot combine because one is a number, the other has x).
>
> **Result:** $32 + 4x$.

1. Simplify the expression: $3 \times 5x + 4x \div 2$.
answer: 17x

2. Simplify the expression: $36y \div 6 + y + 2 \times 4y$.
answer: 13y

3. Simplify the expression: $4n + 3n \times 10$.
answer: 34n

4. Simplify the expression: $32 + 8x \div 2$.
answer: 32 + 4x

5. Simplify: $14pq \times r$.
answer: 14pqr

6. Simplify: $12x^2y \div 3x$.
answer: 4xy

7. Simplify: $7 \times 3a - (3a + 5a) \times 2$.
answer: -16a

Removing and Adding Brackets to Simplify Expressions

> EXPLANATION
> When you have brackets in an algebraic expression you can either remove them by using the **distributive law** or by changing the sign of each term inside the brackets.
>
> - If a bracket is preceded by a **plus sign (+)**, simply drop the brackets and keep the signs of the terms inside.
> - If a bracket is preceded by a **minus sign (–)**, drop the brackets and **reverse** the sign of every term inside.
>
> After the brackets are gone, combine **like terms** (terms that have the same variable raised to the same power). This gives you the simplified expression.

> WORKED_EXAMPLE
> **Worked example**
>
> Simplify \((5b + 3) - (2b - 4)\).
>
> 1. The second bracket is preceded by a minus sign, so change the signs inside:
> \((5b + 3) - (2b - 4) = 5b + 3 - 2b + 4\).
> 2. Remove the brackets (they are now gone):
> \(5b + 3 - 2b + 4\).
> 3. Combine like terms:
> - Variables: \(5b - 2b = 3b\).
> - Constants: \(3 + 4 = 7\).
> 4. Final simplified form: \(3b + 7\).
>
> So, \((5b + 3) - (2b - 4) = 3b + 7\).

1. (a) \((12a - 7) + (4 - 5a)\)
answer: 7a - 3

2. (b) \((8x - 2y) - (9x - 4y)\)
answer: -x + 2y

3. Simplify \(12 - (7 - 2f) - (3 - f)\)
answer: 2 + 3f

4. Simplify \((8a - 5) + (14 - 3a)\)
answer: 5a + 9

Objective 3

Solving Word Problems with Algebraic Expressions

EXPLANATION

When a word problem gives you a relationship between quantities, you can turn the words into an algebraic expression. Follow these steps:

1. Identify the unknowns and assign a letter to each.
2. Write down what you know as an equation or expression using the letters.
3. Use the order of operations (brackets, multiplication/division, addition/subtraction) to simplify the expression.
4. Solve for the unknown by isolating it on one side of the equation.

Remember that "less than" means you subtract, and "more than" means you add.

WORKED_EXAMPLE

Example:

A notebook costs \(n\) naira and a pen costs \(p\) naira. The problem says: "A notebook costs \(\text{₦}5\) more than a pen, and together they cost \(\text{₦}23\)."

Step 1: Let \(p\) be the cost of the pen.
Step 2: Then the notebook costs \(p + 5\).
Step 3: Write the total cost: \(p + (p + 5) = 23\).
Step 4: Simplify: \(2p + 5 = 23\).
Step 5: Subtract 5: \(2p = 18\).
Step 6: Divide by 2: \(p = 9\).
So the pen costs \(\text{₦}9\) and the notebook costs \(9 + 5 = \text{₦}14\).

1. 1. A pen costs \(h\) naira and a crayon costs \(\text{₦}20\) less than a pen. (a) Find the cost of a crayon. (b) Find the total cost of a crayon and a pen.

answer: (a) \(h - 20\) naira (b) \(h + (h - 20) = 2h - 20\) naira

2. 2. Envelopes cost \(d\) naira per pack. Four notepads cost \(\text{₦}20\) more than a pack of envelopes. Find the total cost of one pack of envelopes and one notepad.

answer: Total cost = \(d + \frac{d + 20}{4} = \frac{5d + 20}{4}\) naira

3. A pen and a pencil together cost \(x\) naira. If a pen costs \(y\) naira, find the cost of three pencils.

answer: Cost of three pencils = \(3(x - y)\) naira

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	3	Scope	grounded
Explanation the principle, rule or relationship to review	3	Method written from event labels — the shape of the teaching without its substance	substituted
Worked example a problem with its solution worked through	3	nothing in the input	unsourced
Figure a drawing the learner reads from	0	nothing in the input	not produced
Practice what the learner does	14	Set work	grounded
Answer what the learner checks against	14	Assessment checked against the evaluation answers rather than the taught method	substituted